

■ 1.3.4 Special Topic: Terminating Input Lines

When you type input to *Mathematica*, you have to give some indication of when your input is finished. Exactly what “termination character” you should type depends on the front end you are using.

line-by-line	RETURN
full screen text	CONTROL-X
graphical	SHIFT-RETURN

Typical termination characters used in *Mathematica* front ends.

The *Mathematica* kernel can only process complete *Mathematica* input expressions. Much of the time, your whole input will only be one line long, so that you type the termination character as soon as you get to the end of the line.

If you have a long piece of input, however, you may have to let it continue for several lines. With most *Mathematica* front ends, you can just type RETURN at the end of each line, and then give the termination character when you reach the end of your whole input expression.

When you use *Mathematica* with a line-by-line front end, *Mathematica* will go on reading each successive line you type until it has got a complete expression. This means, for example, that *Mathematica* will go on reading lines of input that you type, until all open parentheses are closed, however many lines this takes. (Note that if you enter a completely blank line, *Mathematica* will throw away your input up to that point, so that you can start over again.)

Mathematica knows that the first line cannot be your whole expression, so it waits for you to give more input on the next line.

```
In[1]:= 3 + 4 +
5
Out[1]= 12
```

Mathematica will wait for as long as is needed to get a complete expression.

```
In[2]:= ( 3
+ 4
+ 5
)
Out[2]= 12
```

If you know that your input will go on for several lines, it is often a useful trick to open parentheses on the first line, then close them only when you get to the very end of your input.